

12.08.2024

ISO 207 - Data Structure & Algo

(1)

$$\Theta(n^{1.6}) \xrightarrow{+} \frac{n \cdot n \log_2^{3/2}}{n \log_2^2, n \log_2^{3/2}} \approx n \log_2^3$$

(2)

Merge (A, n, i, B, m, j, C, k)

$$C[k] \leftarrow \min\{A[i], B[j]\}$$

$k++$

Merge (A, n, i, B, m, j, C, k)

$$T(n) = 2 \cdot T(n/2) + f \cdot n.$$

$$\Downarrow$$

$$\Rightarrow \Theta(n \log n)$$

What if $n \neq 2^k$

$$\lfloor n/2 \rfloor, \lceil n/2 \rceil$$

$$\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ \hline \end{array}$$

1

$$\left(\underbrace{\dots}_{\substack{\text{for } \dots}} \right)$$

Fact $\forall x \in S \exists$ perfect power of 2 $\in [x, 2x]$
8 $\in [5, 10]$

Median Finding

Defⁿ [i-th Statistic]

i-th smallest element in the array.

Defⁿ [Median]

$\lfloor n/2 \rfloor$ -th statistic [lower median]
 $\lceil n/2 \rceil$ [higher median]

Q: Given array A, return the i -th smallest.

A: Sort low to high & return i -th one.

$O(n \log n)$ time

Q: Can we do better?

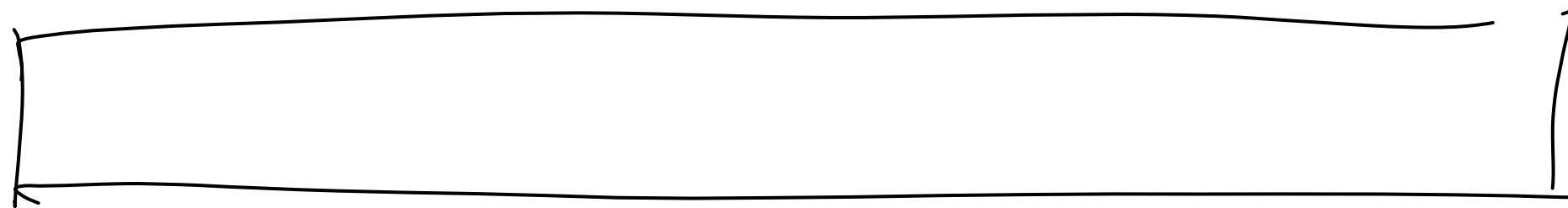
Ans $O(n)$ time!

Suppose $i = 1, 2, \dots$, constant.

$$i = \lfloor n/2 \rfloor$$

i -th order statistic

input



$$T(n) = a T(\frac{n}{b}) + Cn$$

~~3~~ 2 $\rightarrow n^{\log_2 3}$

$$a = 2, b = 2 \rightarrow n \log n$$

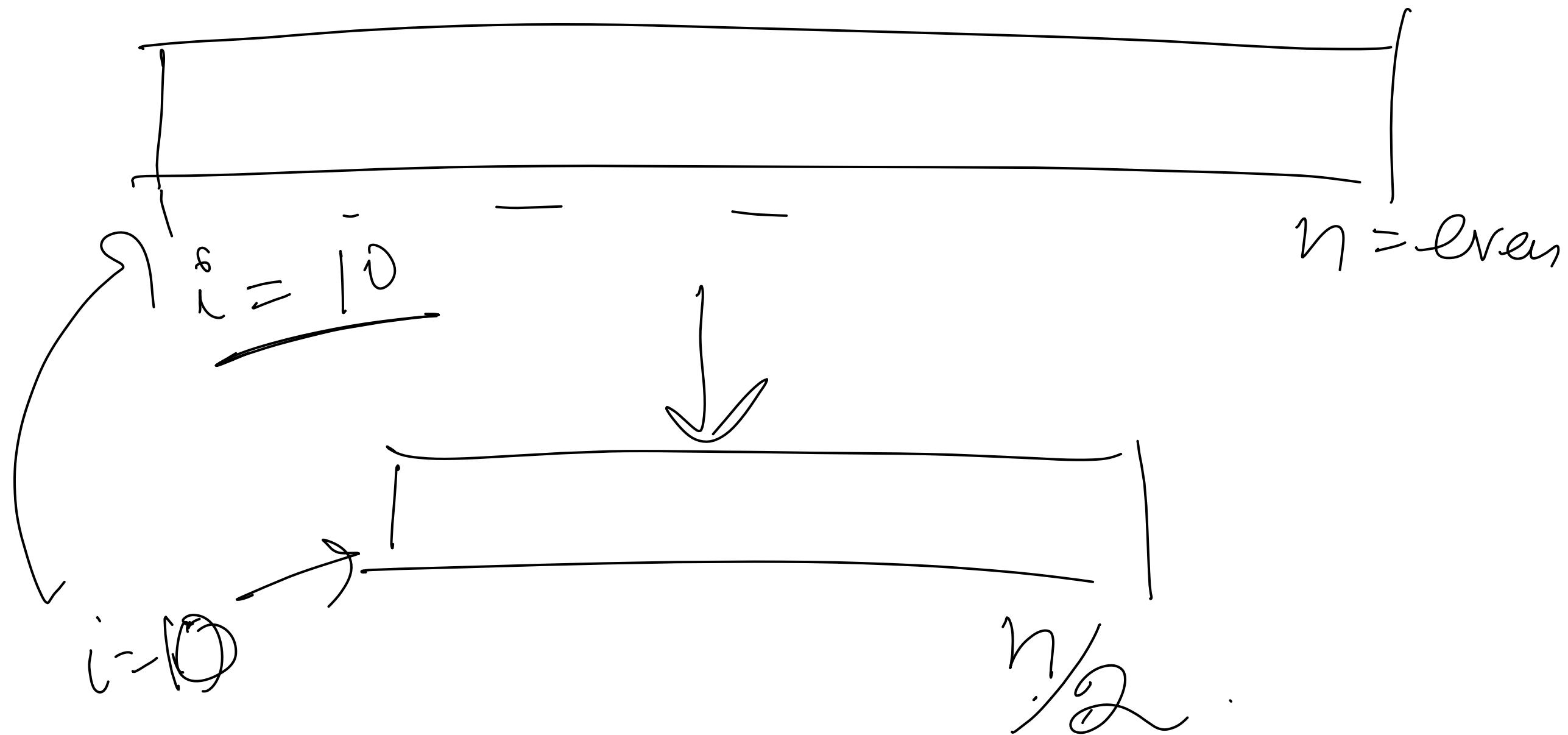
$$= a = 1; b = 1$$

$$T(n) = T(0.9n) + cn$$

~~$$T(n) = T(0.9n) + cn$$~~

$$T(n) = \Theta(n)$$

A 0



need?

Some element of the array whose rank
is between 25% to 75%
~~10%~~ ~~90%~~

Smaller

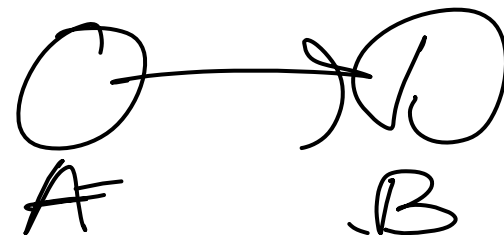
$n/4$

larger

$n \text{ and } 5 \approx 0$

median

Sort



$A > B$

High-level pseudocode

(1) putting all items in a $(\overline{5} \times g)$ -grid

$$g = \frac{n}{5}$$

(2) sort each of the g columns.

(3)

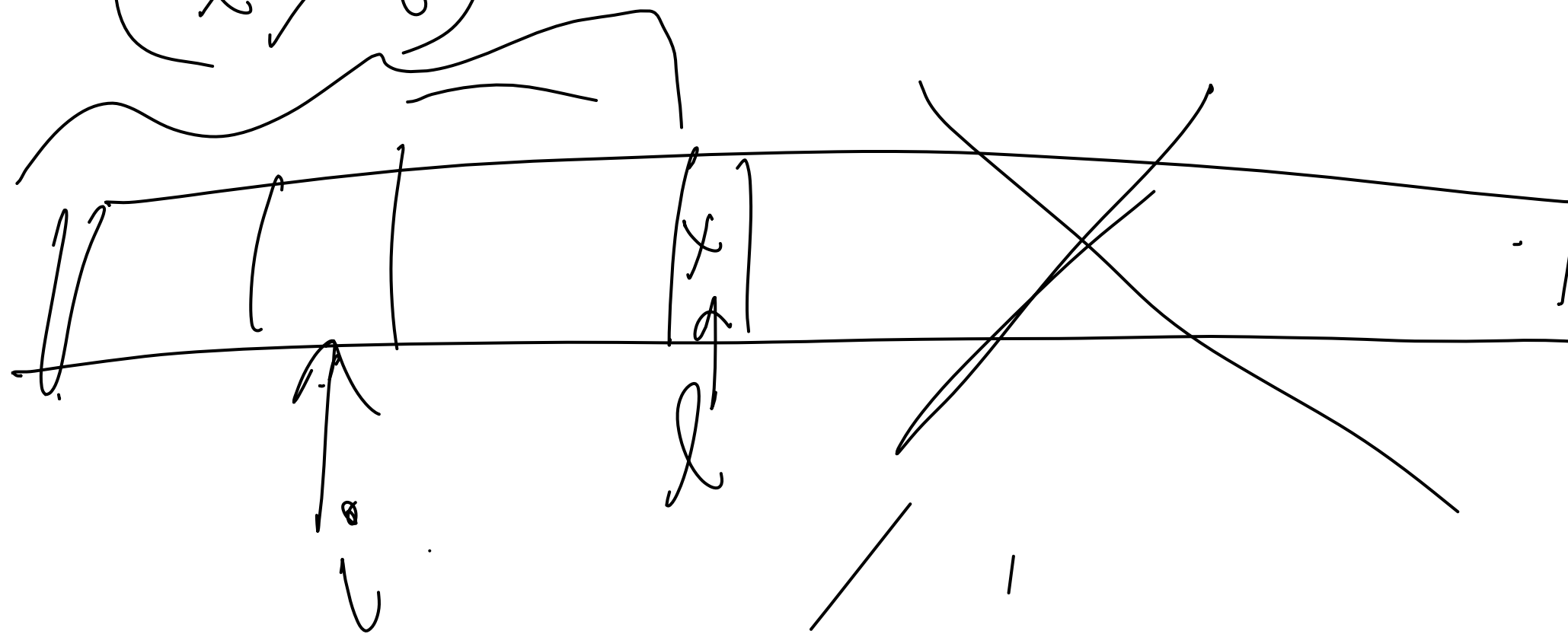
~~ca~~ find median of middle row $\rightarrow \times$

(4) find out rank of $x \leftarrow l$

(5) if $(l \leq i)$ return ~~x~~.

(6) ^{or} if $(l > i)$

$A[i]$
(sorted)



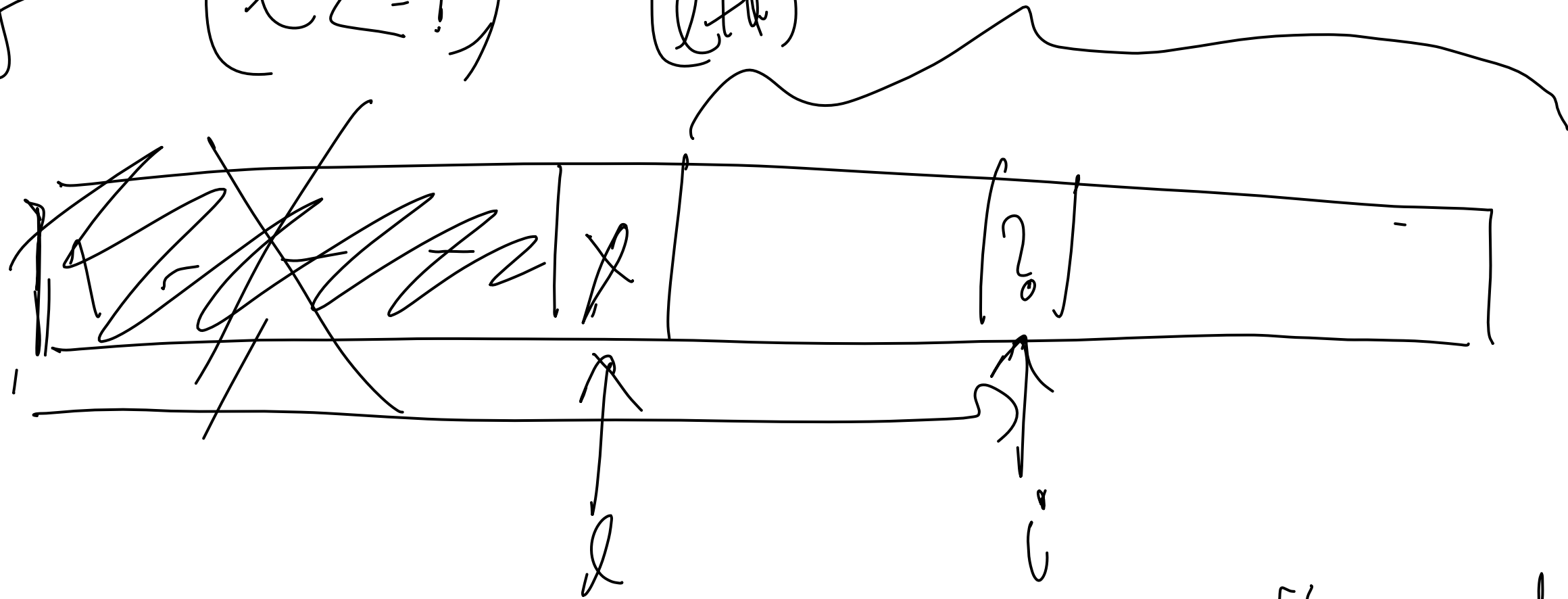
~~find~~
Create array B of all items with
rank $< l$

Call i th statistic on B

(7)

if $(l < i)$ $(l+1)$

A
(sorted)



Create an array B with all items having

$\text{rank} > l$

Call $(i-l)$ th orderstat on B .

Two recursive calls

- median : $n/5$ items
- other : $\leq 3n/4$ items.

$$T(n) \leq T(n/5) + T(3n/4) + Cn.$$

~~is~~
 $0.2n$

$0.75n$

$$T(\underbrace{0.95n}_{< n})$$

$$O(n)$$

Procedure OrderStat (A, n, i)

$h \leftarrow n \bmod 5$

for $p = 1$ to $h \bmod 5$

// Assume 1 index.

for $j = p+1$ to n

if ($A[p] > A[j]$)

// swap $A[p]$ & $A[j]$

end if

end for

// Create a new array B with all
// first $n \bmod 5$ items of A.

--- $m \leftarrow \text{sizeof}(B)$

// make ~~g~~ $g \leftarrow m/5$ good. C.
C $\leftarrow$ empty ~~g~~ g ~~AS~~ array.

~~k~~ $k \leftarrow 1$
 $i \leftarrow i - (n \bmod 5)$

for $p = 1$ to g .

for $j = 1$ to 5

$C[p][j] \leftarrow B[K]$

$K++$

end for

end for.

~~Sort~~ find median of $C[3]$ —

~~#~~ ~~D~~

// create a new array D.

// D[i] will have median of

// $C[1][i], C[2][i], C[3][i], C[4][i], C[5][i]$

D ←

$X \leftarrow \text{OrderStat}(D, \cancel{L}, \lceil G/2 \rceil)$

count $\leftarrow 0$

for $p = 1$ to m

if $(B[p] < X)$

count $\leftarrow +1$

end if

end

$l \leftarrow \text{count} + 1$ // rank of X

u

~~*~~ if ($i > l$)

~~F~~ $F \leftarrow$ empty array of size ~~($m-l$)~~

$K \leftarrow (m-l)$ items.

$K \leftarrow 1$
for ($j = 1$ to $(m-l)$)

if ($B[j] > x$)

$F[K] \leftarrow B[j]$

$K++$

return ~~F~~

Order Stat. ($F, m-l, i-l$)

if (i < l)

!

if (i = l)

return X

